

How Quality Data Sets Help Machines Learn

Just like the fuel that runs the engine of a vehicle, a data set is a key ingredient for solving a given problem using machine learning or deep learning algorithms. This article takes a quick look at open source data sets for ML and DL, with a focus on TensorFlow and PyTorch.

There is a lot of research going on in and a lot of applications being built with machine learning. Machine learning (ML) algorithms automatically transform data into useful representations for the task in hand. These operations could be co-ordinate changes, linear projections (linear regression), translations (SVM), transformations (PCA), and so on. These algorithms are not usually creative. They just search in a hypothesis space.

Deep learning (DL) is a specific field of machine learning which emphasises learning in successive layers to get more meaningful representations of data. Deep in deep learning represents the number of layers or the depth of the layers used in it.

There are two kinds of problems in machine learning — regression and classification. Just suppose we are required to predict the cost of a

house. Here, we need to get a numeric value. This becomes a regression problem. However, sometimes we need to take a “Yes” or “No” decision. This is a binary classification problem. At other times, we need to predict a class. Consider a problem where we are required to identify an animal like a cat, dog, horse, etc. This is a multi-class classification problem.

Whenever we get the data for a problem for supervised learning, we get attributes and labels. Attributes are independent variables with which we must predict the labels or dependent variables. Attributes are called inputs, predictors, features, or independent variables. Labels are called outputs, targets, outcomes, or dependent variables. If the labels are categorical, the problem becomes a classification problem; if these are numerical, it becomes a regression problem.

Importance of data sets for ML and DL

Machine learning or deep learning tries to learn patterns from the data available using algorithms. Data is the core of any ML/AI algorithm. Training data for ML is a key input to the algorithm that learns from it, and memorises the information for future prediction. Training data is the backbone of the entire machine learning project, without which it is not possible to train a machine that learns from humans. Raw data collected from various sources needs to be labelled or annotated so that it can be used to predict in the case of supervised machine learning.

Availability of such annotated data is very important to train and validate the algorithms. The quality and size of the data set is very important for training of the learning method in order to obtain good results and high performing models. Raw

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